

| Question |     |  |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Marks available |          |          |          |          |          |
|----------|-----|--|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|----------|----------|----------|
|          |     |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | AO1             | AO2      | AO3      | Total    | Maths    | Prac     |
|          | (c) |  |  | <p><b>A</b> ammonium chloride / <math>\text{NH}_4\text{Cl}</math></p> <p><b>B</b> copper(II) sulfate / <math>\text{CuSO}_4</math> accept copper sulfate</p> <p><b>C</b> potassium carbonate / <math>\text{K}_2\text{CO}_3</math><br/>accept potassium hydrogencarbonate / <math>\text{KHCO}_3</math></p> <p>award (1) for each compound correctly identified</p> <p>if all compounds not correctly identified<br/>award (2) for any four ions identified<br/>award (1) for any two ions identified</p> <p>accept ions correctly identified in cases where incorrect formulae are given</p> | 3               |          |          | 3        |          | 3        |
|          |     |  |  | <b>Question 8 total</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>3</b>        | <b>3</b> | <b>2</b> | <b>8</b> | <b>0</b> | <b>3</b> |